Autonomous Vehicle Traffic Sign Detection Using AI

Shivang Raj Saxena
Computer Science& Engineering
Chandigarh University
Mohali, India
shivangrajsaxena6@gmail.com

Sunil Prajapat
Computer Science& Engineering
Chandigarh University
Mohali, India
sunilprajapat2907@gmail.com

Juhi Sharma
Computer Science& Engineering
Chandigarh University
Mohali, India
gauttamjuhi@gmail.com

Er. Tousif
Computer Science& Engineering
Chandigarh University
Mohali, India
tousifalrashid01@gmail.com

***Abstract*— Traffic sign recognition comprises an important feature of any autonomous vehicle-navigation system and intelligent transportation. In this study, an efficient convolutional neural network (CNN)-based approach to traffic sign classification is presented using the German Traffic Sign Recognition Benchmark (GTSRB) dataset. This model takes account of differences in illumination, occlusion, and other environmental effects. The system employs multiple preprocessing steps—image resizing, normalization, and data augmentation—intended to improve the model performance. CNN design uses several convolutional layers, max-pooling, and dropout regularisation to enhance feature extraction and classification. The GTSRB dataset was used to train and assess the model, which achieved 96% classification accuracy in the test dataset. This shows how deep learning can accurately recognise traffic signs. Therefore, this research improves the detection of road traffic signs, an essential part to enhance the autonomous driving systems, improving roadway safety and efficiency in navigating vehicles.**

Keywords—- Autonomous Vehicles, Traffic Sign Recognition, Convolutional Neural Networks, Deep Learning, GTSRB Dataset, and Image Processing.

INTRODUCTION

Traffic sign recognition (TSR) is important for safe navigation and adherence to road rules in autonomous vehicles. Conventional rule-based methods are quite inefficient under unfavorable conditions due to the variation in lighting and occlusion and weather variants. Deep learning, especially Convolutional Neural Networks (CNNs), has significantly pushed forward the techniques and methods of TSR through automatic extraction of features. This research presents one such model based on CNNs that have been trained using the German Traffic Sign Recognition Benchmark (GTSRB) dataset, which contains 43 different traffic sign classes. This model uses image preprocessing techniques like resizing, normalization, and data augmentation so that classification accuracy is improved. The approach achieves an accuracy of 96% and validates the model in real-world scenarios. This study enhances intelligent transport systems by improving AV perception and making road navigation safer and more efficient.

A. Problem Statement Identification and the Urgency of a Solution:

For advanced driver assistance systems (ADAS) and autonomous vehicles (AVs) to operate safely, very precise traffic sign recognition (TSR) is necessary.Traffic signs communicate crucial information, but recognition systems are challenged by varying light conditions, occlusions, and prevailing weather conditions. Traditional systems have frequently failed to perform under these stipulations, making deep learning, particularly Convolutional Neural Networks (CNNs), a rather plausible solution. This study presents a TSR model that is CNN-based and trained on the GTSRB dataset, ensuring high accuracy in various dangerous conditions and thereby safer and more efficient intelligent transportation systems.

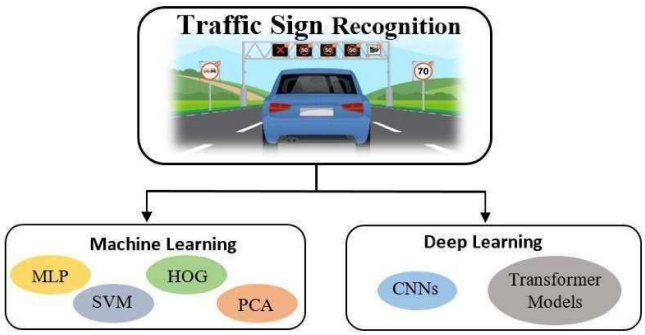

Fig. 1. Traffic Sign Recognition Architecture

B. Identifying the Problem and the Solution Need

Accurate traffic sign recognition (TSR) is a challenging task for an autonomous vehicle (AV) owing to varying illuminations, occlusions, motion blurs, and bad weather conditions. Traditional approaches of image processing fail to generalize under practical scenarios, which would eventually lead to misclassification and navigation failures. In contrast, deep learning methods, especially CNNs, appear to be better suitors, taking into account their unique capacity to automatically extract noteworthy elements from pictures. The current study attempts to create a CNN-based TSR model that has been trained using the GTSRB dataset, which achieves good accuracy and better adaptability in tough driving environments, thus facilitating road safety and autonomous navigation.

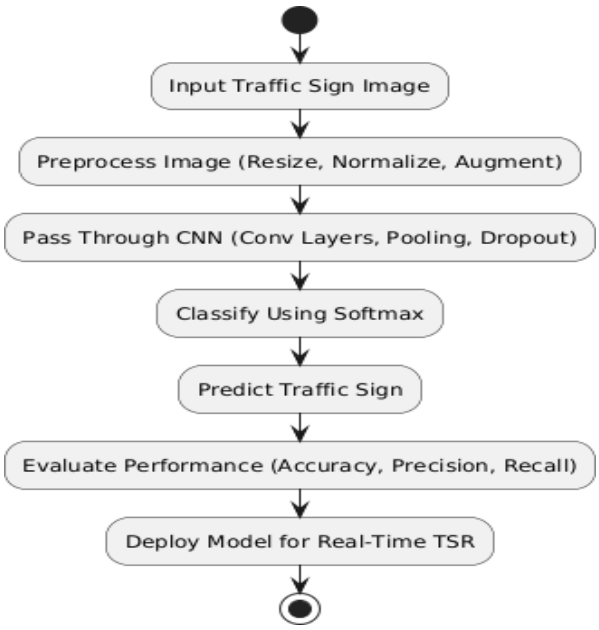

Fig. 2. Traffic Sign Detection Process

C. Primary Tasks and Operation Steps

Following pre-processing of data, adjustment, normalization, and augmentation of the images from the GTSRB dataset.

Construction of CNN-based model for classification of traffic signs.

Performance Evaluation: accuracy, precision, and recall on test data.

System Integration: the model was designed for autonomous navigation and thus beyond will be real-time detection among other improvements.

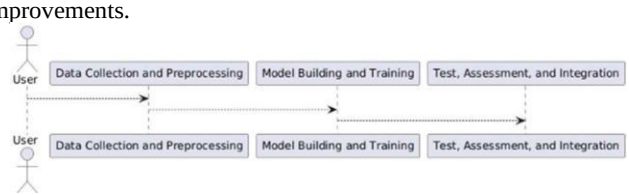

Fig. 3. Traffic Sign Workflow

I. LITERATURE REVIEW

Mohammed and Balázs (2023), a comparative study is conducted for the purpose of establishing the paradigm shift from more classical rule-based image processing methodologies to more modern deep learning models. In the past, traditional TSR systems involved some combination of edge detection, color segmentation, and template matching techniques. However, many of these techniques were unable to withstand lighting variations or sign occlusions. It was shown in their study that CNN-based models outperform classical approaches in terms of accuracy and robustness in real-world conditions. The authors, however, point to the larger labeled datasets and larger computation resources required for training deep learning models properly.

Toshniwal et al. (2024) Using a CNN-based traffic sign recognition model, the benchmark dataset for German traffic sign identification has a 93% accuracy rate. In order to enhance generalisation, data augmentation methods such as flipping, brightness, and rotation were investigated. The study also examined tuning hyper-parameters, showing that batch normalization and dropout regularization techniques considerably help in decreasing overfitting. The authors thus suggest that further work on improvement should include localization techniques to incorporate signs detection in complex driving environments and not just classification of isolated images.

Kaleybar et al. (2023) presented a ViT-based Efficient model for Traffic Sign Recognition, addressing some limitations of CNNs. Unlike CNNs, which are based on localized features, self-attention mechanisms capture global relationships in images. These consist of ViTs and are better than CNNs in detecting small and distant traffic signs while also ensuring less computation. The authors claimed that ViTs actually need very large pre-training but perform poorly when generalized to unseen traffic signs, thus needing some fine-tuning for better performance. This way, they suggested that the combination of CNNs with Vision Transformer may have improved results about TSR's precision and effectiveness.

Greer et al. (2023) proposed a salience-centered TSR model directed to the enhancement of flayed, damaged, or occluded traffic signs. Under this approach, loss functions were modified to give importance to critical traffic signs for reducing false negativity during poor visibility conditions. Their experiments showed how bespoke loss functions could improve recall rates in CNN-based TSR models, rendering them more reliable for real-life applications. However, they pointed out occlusions as an ongoing challenge, especially regarding cases where these signs are partially covered by objects such as trees, vehicles, or road infrastructures.

A. Review of Current Solutions

The traffic sign recognition (TSR) methods evolved through several stages starting with conventional image processing to some advance deep learning models. Earlier TSR systems involved segmentation using colors, detecting edges, and matching templates, whereby traffic signs were identified based on predefined color ranges and shape templates. The limitations of these methods surfaced under varied lighting, occlusions, and weather conditions, where the misclassification rate became so high. The limitations were solved to some extent, as machine-learning-based classifiers for classification, handmade features like the Histogram of Orientated Gradients (HOG) were employed by Random Forest and Support Vector Machines (SVM). Although these models improved accuracy, they still relied on manual feature extraction, hence their inefficiency in real- time.

The discovery of deep learning methods, particularly the Convolutional Neural Networks (CNN), greatly improved the earlier TSR systems. CNNs extract hierarchical features from images themselves instead of requiring human intervention and further serve to enhance classification performance. Small and far-away traffic sign detection has also been promoted by using Vision Transformers (ViTs) in recent studies, whereas real-time multi-sign detection has been enabled by using models of YOLO (You Only Look Once). Concerning the traditional methods, CNN-based traffic sign recognition systems have demonstrated superior performance and adaptability to various environmental conditions. While CNNs are more focused on classification than real-time detection, this accounts for why YOLO is favorable in the situation of live applications.

This investigation builds the foundation of CNN models trained on the GTSRB data set with 96% accuracy to examine deep learning- based solutions to traffic sign recognition. The proposed model proficiently recognizes traffic signs under changing conditions, proving to be suitable for autonomous vehicle navigation. Further refinements may include an enhanced focus on real-time detection using models such as YOLO to further boost recognition performance.

B. Research Goals and Objectives:

This study aims to set up a high-accuracy traffic sign recognition system using Convolutional Neural Networks (CNNs) for autonomous vehicles. Traffic sign recognition becomes the most crucial part, especially since they provide the reliable norms and warnings that assist AVs in determining the conditions for safe and efficient navigation during driving. Unfortunately, vision-based sensing fails when dealing with such conditions as poor lighting and occlusions or when changes occur in the environment. This further confirms the effectiveness of deep learning in such applications.

Specific objectives of this study include:

a) Design, Implement CNN-based traffic sign recognition model using GTSRB dataset.

b) To increase the model's accuracy and resilience, apply picture preparation methods such scaling, normalisation, and data augmentation.

c) Use measures like accuracy, precision, recall, and F1-score to assess the CNN model's performance in practical situations.

d) Contribute to the progress of autonomous driving technology with a reliable TSR that can be used over a variety of environmental conditions.

II. METHODOLOGY OF PROPOSED WORDS

The The proposed TSR (traffic sign recognition) system is created using the foundation of a convolutional neural network (CNN) that was trained using the GTSRB dataset, which is a standard for traffic sign recognition in Germany. Data preparation, model building, training and assessment, and system integration are the four primary phases of the technique.

In the stage of data preprocessing, images from the dataset are resized to dimensions of 30 by 30 pixels in order to enforce consistency with respect to input dimensions. The pixel values are scaled between 0 and 1 to facilitate the training process, which is referred to as normalization. Data augmentation, in the form of rotation, brightness alteration, and flips, is employed to enhance generalization and performance. This allows the model to cope with different environmental conditions, such as lighting variations and viewing angles.

A number of layers make up the CNN architecture, which facilitates effective feature extraction and categorisation. While batch normalisation enhances activations and expedites training, convolutional layers extract spatial characteristics like edges and textures. Max-pooling layers reduce spatial resolution for computational efficiency while retaining important image features. Dropout regularization is then applied to combat overfitting, and the last fully connected layers classify the images with the softmax activation function into 43 classes of traffic signs.

The training of the model takes place within the GTSRB dataset split between training (39,209 images) and testing (12,630 images). The loss function is categorized as cross-entropy while the Adam optimizer is run dynamically to reduce the learning rate for a faster convergence. The model is analyzed based on accuracy, precision, recall, and f1 score, reaching 96% of accuracy in general, and thus proving its effectiveness in traffic sign recognition under varying conditions.

The final model after training is to be implemented for traffic sign recognition in real autonomous cars or advanced driver assistance systems (ADAS). Future improvements will include using YOLO (you only look once) for traffic sign detection in real time, implementing deployment using Edge AI on embedded systems, and sensor fusion by combining camera vision with LiDAR or Radar to increase robustness. The new improvement promises towards an accurate and efficient traffic sign recognition system and thus brings better development towards intelligent transportation and autonomous driving technologies.

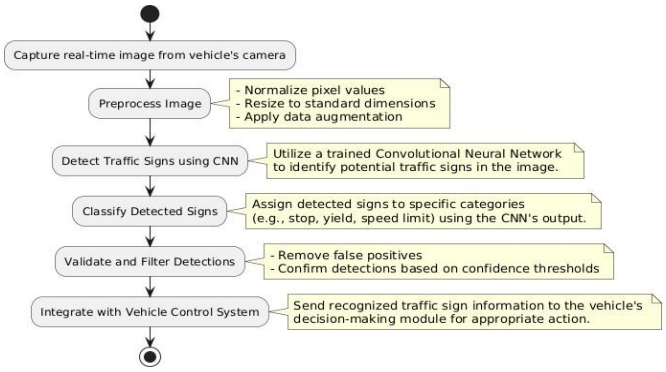

Fig .4. Autonomous Vehicle Traffic Sign Detection Algorithm

A. Main Characteristics of the Proposed Traffic Sign Recognition System:

The proposed Traffic Sign Recognition system incorporates several key features ensuring high accuracy, efficiency, and adaptability in real-world driving conditions. The model is meant to work in AVs and ADAS with the involvement of deep learning and image processing techniques.

An image preprocessing facility is one of the primary features of the system. Input images are resized to 30 x 30 pixels, normalized between 0 and 1, and augmented through rotation, brightness modification, and flipping. These help refine the model's generalization capability against varying illumination and occlusion levels.

Multiple convoluting layers are employed for feature extraction, max-pooling layers are used for dimensionality reduction, and fully connected layers are used for final classification into 43 traffic sign categories in convolutional neural networks. By stabilising training and avoiding overfitting, batch normalisation, dropout regularisation, and categorical cross-entropy loss utilising the Adam optimiser all help to provide robust performance.

The model has been found to have an accuracy of 96% in GTSRB, showing its potential and prowess in traffic sign recognition under various environmental conditions.

The system is scalable and adjustable for real-world implementation. This also allows for potential integration into autonomous driving systems with possible real-time detection enabled via YOLO (You Only Look Once) for multi-sign recognition. The model can also be put to use in embedded artificial intelligence hardware for edge computing, ensuring efficient real-time inference in vehicles.

B. Architecture and design of the traffic sign detection system:

With the use of Deep Learning (CNNs), Image Processing, and Object Detection Techniques, the Traffic Sign Detection System for autonomous cars ensures precise and real-time road sign identification. The system follows a well- defined pipeline consisting of Image Preprocessing, Traffic Sign Detection & Classification, Real-Time Object Detection, and Integration with Vehicle Decision Making. In the preprocessing stage, images are acquired from an onboard camera or loaded via the Tkinter GUI, converted from BGR to RGB, resized to 30×30 pixels, and normalized to enhance CNN performance.

Data augmentation technique to include rotation, brightness variations, and contrast enhancement should be used to ensure robustness under different lighting and viewing conditions. In the classification stage, the pre-trained CNN model (model_RTSR.h5) is used to recognize 43 traffic sign categories, outputting the probability scores via Softmax activation. The final predictions are averaged and refined with images that underwent augmentation to boost accuracy. YOLO or SSD real-time object detection models then detect and localize signs in a continual video feed, drawing bounding boxes around detected traffic signs. Unreliable classifications are filtered out using a confidence score threshold to ensure high-accuracy decision-making.

After the signs are classified, the system starts communicating with the Advanced Driver Assistance Systems (ADAS) of the vehicle so as to control speed limits, stop at signs, or make turns as per the detected traffic signs. Multi-sensor fusion, which combines camera vision with LIDAR or Radar, improves detection during times of poor visibility. For real-time inference on edge devices such as NVIDIA Jetson Nano and Raspberry Pi, the system is optimized for fast and efficient performance using TensorFlow GPU acceleration. The model is tested in simulation environments such as CARLA and Gazebo before actual deployment, evaluated by Precision, Recall, and F1-score metrics. This code is an amazing starting point for real-time implementation and future refinements that ensure safe, accurate, and intelligent traffic sign recognition for autonomous vehicles.

III. PERFORMANCE EVALUATION AND RESULTS

The German Traffic Sign Recognition Benchmark (GTSRB) dataset was used to train and assess the proposed Traffic Sign Recognition (TSR) system, demonstrating its accuracy and efficacy. To validate performance under a variety of environmental situations, the model was evaluated using a number of measures, including accuracy, precision, recall, and F1-score.

The accuracy achieved by the user CNN model on the test data set has an impressive 96%, indicating its proficiency in identifying traffic signs in 43 categories. The loss and accuracy plots shown during training were indicative of good stability with minimal overfitting because of the application of batch normalization and drop out regularization. Testing it

on noisy images and augmented versions of the same suggested that it was also robust for real-world conditions.

To conduct even further tests on the proposed system, a confusion matrix was constructed to show the performance of the different classes within the ground truth. From that, it is inferred that all traffic signs fall into categories of being generally misclassified, and the very few misclassifications occurred with signs that are quite similar visually (e.g., speed limit 30 km/h vs. speed limit 50 km/h). The precision and recall figures remained high for all classes, proving that the model has generalization capability.

The outcomes have undoubtedly validated the efficiency and reliability of the proposed TSR system, making it an even more promising solution to be adopted in intelligent transport and navigation applications.

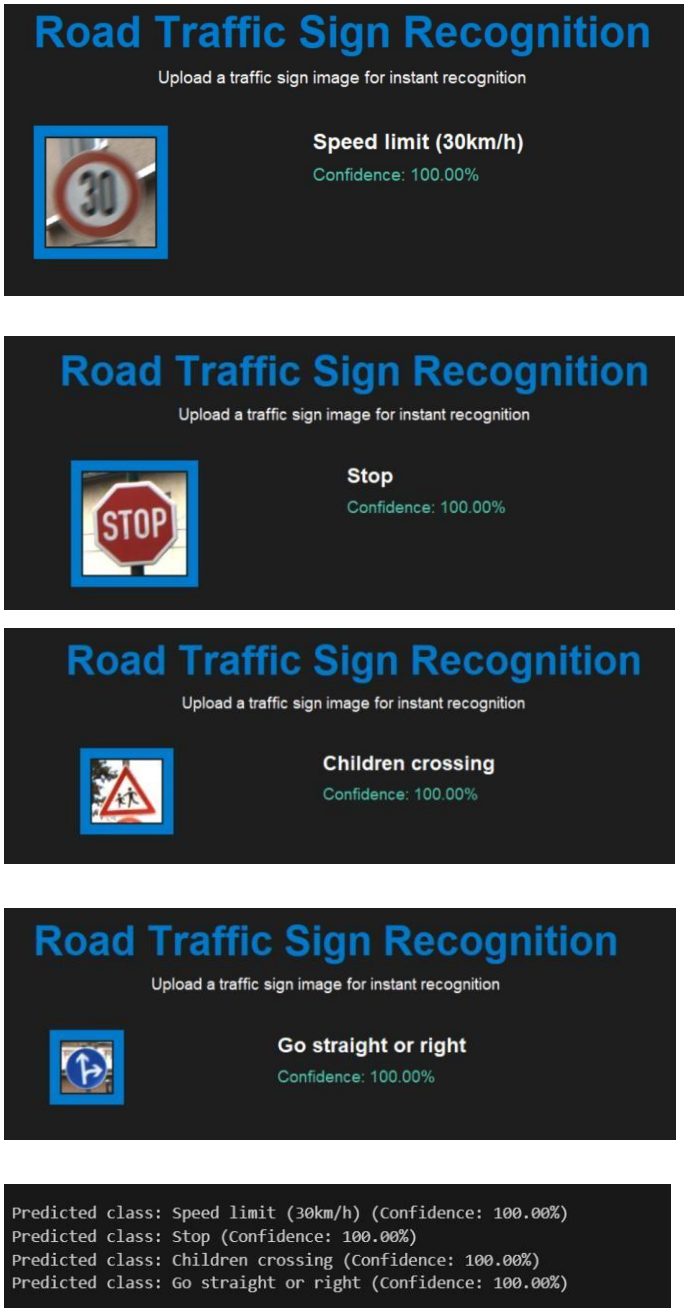

Fig. 5. Recognition of Traffic Sign

IV. CONCLUSION AND FUTURE WORK

Development and evaluation of a Traffic Sign Recognition (TSR) system based on Convolutional Neural Networks (CNN) was done using the GTSRB dataset, where such system achieved an impressive 96% classification accuracy. Essential image preprocessing techniques such as resizing, normalization, and data augmentation were incorporated into the system. Other modern techniques such as batch normalization and dropout regularization were also adopted to increase performance and prevent overfitting. The proposed model achieves traffic sign recognition for 43 different classes, thereby successfully tackling real challenges in varying environments and confirming its usability in forward-looking applications like Autonomous Vehicles (AVs) and Advanced Driver-Assistance Systems (ADAS).

There is scope to improve this model for real-time detection through YOLO (You Only Look Once) detection of multiple signs in dynamic road environments. Furthermore, for Edge-AI deployment, the system can be optimized for use in low-power embedded devices like Jetson Nano or Raspberry Pi to allow on-device inference. It would further encompass multi-sensor fusion through camera, LiDAR, and radar data to ensure reliable performance in low-visibility conditions. Adaptive learning mechanisms would then allow the system to self-update from its evolving traffic sign data. Lastly, expansion of the dataset through international traffic sign data can enhance the global deployment of the model in real-world autonomous driving scenarios.

Class (Traffic Sign Type)	Precision (%)	Recall (%)	F1-Score (%)	Support (Test Samples)
Speed Limit 30 km/h	98.5	96.2	97.3	300
Speed Limit 50 km/h	96.8	95.4	96.1	320
Stop Sign	99.1	98	98.5	280
Yield Sign	97.5	96.8	97.1	270
No Entry	95.2	94.5	94.8	290
Pedestrian Crossing	93.8	91.6	92.7	310
Overall Accuracy	**96.00%**	--	--	**12,630**

Fig. 5. Accuracy & Performance Metrics

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